

Yuchen Wang

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Education

University of Illinois Urbana-Champaign , Computer Science	Champaign, IL, USA 2025 – 2027
Peking University , Intelligent Science and Technology (Artificial Intelligence) • Zhi Class (Elite Program), Top 10%	Beijing, China 2021 – 2025

Experience

Alibaba U.S. , Research Intern — Agentic LLM Post-Training - Core contributor to Occamy-1.0, a 35A3B co-work model. Architected and delivered execution-grounded data and post-training infrastructure for long-horizon tool use; the team's recipe combines SFT, HDPO, expert merging, and SAO over a 15K-trajectory, 403M-token SFT corpus. - Built verifier-gated pipelines for OpenClaw, Hermes, and Accio Work: token-exact replay, state recovery, and compaction-aware segments preserving tokens, log-probs, and loss masks for SFT/RL. - Released model improves Claw-Eval from 69.5 to 82.2 (71.4 Pass³) and AutomationBench from 7.5% to 27.6%; scores 49.16 on WildClawBench and 59.0 on Terminal-Bench 2.1. Reaches the low-cost knee of the cost-performance Pareto frontier in the report's four-benchmark comparison.	Sunnyvale, CA, USA May 2026 – Dec 2026 8 months
Freedo Technology , Research Intern — 3D Vision and Generative AI Built an end-to-end depth-conditioned ControlNet pipeline for 3D reconstruction from noisy point clouds, improving geometric fidelity from 32.7% to 85.3%.	Beijing, China May 2025 – Aug 2025 4 months

Open Source Projects

Open-Source LLM Systems Engineering

Upstream reliability and performance work across FlashInfer, vLLM, NVIDIA NeMo, Megatron-LM, Hermes Agent, SGLang, verl, vLLM/Vime, and ModelScope.

- Numerics and training: FlashInfer #4984 fixes FP8 KV prefill calibration. Contributions also cover Muon and hybrid Mamba/MoE correctness, sequence packing, NCCL startup, non-finite RL logprobs, GDN/Muon clipping and routing, skipped-tensor weight checks, and FSDP validation.
- Performance and scale: vLLM #54699 halved TP2 peak allocation from 7.88 to 3.94 GiB and enabled real 120B load, generation, and 3/3 sleep-wake cycles in earlier validation; NeMo RL #3943 delivered 4.44–5.31× controlled Ray transfer speedup with 0.1–0.3 MB driver RSS deltas; Megatron-LM work fused GatedDeltaNet normalization and added Mamba recompute.
- Agent runtime: Hermes #100693 and #102549 repair schema-reference argument coercion and ownership-checked SSH backend cleanup (22 and 14 targeted tests passed). Co-authored NeMo Gym #2726 for pickle-safe cross-process failures; SGLang #38063 adds cold-JIT startup diagnostics.

Projects

CineFlow — Dependency-Driven Parallel Video Generation

The first dependency-driven video-diffusion inference system, built around semantic DAGs, runtime trajectory fusion, online edge pruning, and critical-path scheduling without retraining.

- Analyzed 1K+ MovieGenBench prompts; more than 68% exhibit non-linear dependencies.
- Across Wan2.2-5B, CogVideoX-5B, and HunyuanVideo on 8×H100, achieved 1.7–5.5× end-to-end speedup, 1.30–2.02× lower P90 latency, and 5.4–17.3% higher VBench overall; pruning added nearly 7% throughput and improved visual coherence by 6.7–14.6%.

Dynamic Prefill Optimization via Adaptive Online Packing

AIMD control with live p95 TTFT feedback and greedy/DP prompt packing.

- Achieved up to 20% lower TTFT on production-style DynamoLLM traces.

FlashAttention-style CUDA Optimization for GPT-2

Tiled online softmax, memory-efficient attention, and kernel fusion.

- Reduced HBM traffic by roughly 10× and achieved up to 9% end-to-end speedup.

Reinforcement Learning for Legal Reasoning

Advised by Prof. Yansong Feng at Peking University; Zero-RL → distilled-CoT SFT → GRPO for structured legal multiple-choice reasoning.

- Reached 57.6% accuracy, outperforming SFT-only and RL-only baselines.

Skills

- Python, C++, CUDA, PyTorch, Ray, FSDP, 3D vision, point clouds, ControlNet, diffusion transformers, Megatron-LM, NeMo RL, vLLM, SGLang, distributed training, and profiling